

# INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

Department of Computer Science and Engineering

## **BTP Final Report**

### **Diabetes Risk Prediction using Explainable Artificial Intelligence**

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## **Abstract**

This report presents a complete, reproducible, and interpretable machine learning pipeline for early diabetes risk prediction using the Pima Indians Diabetes Dataset. The workflow spans data quality checks, robust preprocessing, baseline model training (Logistic Regression, Random Forest, XGBoost, LightGBM), multi-metric evaluation, and explainable AI using SHAP to provide global and local transparency.

The system is packaged with automated reporting (PDF and HTML dashboard) and a deployable user interface (Streamlit/Gradio) for real-time inference. Experiments reveal that glucose, BMI, age, and pregnancies consistently emerge as the most influential predictors. The pipeline emphasizes not only performance but also accountability, enabling end users to inspect how each feature contributes to a prediction.

# Chapter 1

## Introduction

### 1.1 Background

Diabetes mellitus is a chronic disease characterized by elevated blood glucose levels that can result in severe long-term complications. Large-scale screening and early risk stratification are essential to reduce morbidity. Machine learning (ML) methods offer a data-driven approach for early warning systems. However, opaque models limit clinical trust and adoption. This work integrates ML with explainable AI (XAI) to deliver both predictive strength and transparent reasoning.

### 1.2 Motivation

Healthcare stakeholders increasingly demand interpretable models. Feature attributions that are faithful, stable, and human-readable help clinicians calibrate trust and spot spurious correlations. SHAP (SHapley Additive exPlanations) fits these requirements by providing consistent, additive explanations grounded in cooperative game theory.

### 1.3 Problem Statement

Design an end-to-end ML pipeline that (i) cleans and preprocesses the Pima dataset; (ii) trains multiple algorithms; (iii) evaluates models using robust metrics; (iv) provides global and local explanations using SHAP; and (v) exposes an interactive interface (Streamlit/Gradio) for live predictions and interpretability.

## 1.4 Contributions

- A reproducible preprocessing pipeline with safe handling of biologically impossible zeros and leakage-free imputation.
- Comparative baselines across four classifiers with multi-metric evaluation.
- Comprehensive SHAP analysis: beeswarm, mean |SHAP|, dependency plots, and individual waterfalls.
- Automated report generation (PDF) and lightweight HTML dashboard.
- Deployable UI (Streamlit/Gradio) that accepts patient features and returns predictions with explanations.



# Chapter 2

## Literature Review

### 2.1 Diabetes Prediction

The Pima Indians Diabetes dataset is a standard benchmark for binary classification on tabular medical data. Early approaches emphasized logistic regression and decision trees; recent studies show gradient-boosted trees and random forests improve recall and F1 by capturing nonlinear interactions among clinical variables such as glucose, BMI, age, and insulin.

### 2.2 Explainable AI in Medicine

Model transparency is crucial in clinical workflows. Post-hoc techniques such as LIME and SHAP are popular; SHAP provides local and global attributions that satisfy additivity and consistency, qualities that aid auditing and increase clinician confidence. Prior work highlights SHAP’s ability to uncover influential factors and validate medical intuition.

### 2.3 Gaps Addressed

Many pipelines optimize accuracy but omit interpretability, report automation, and deployment. This work integrates the full stack—data quality, modeling, XAI, reporting, and UI—yielding a turn-key demonstrator suitable for evaluation and future clinical prototyping.

# Chapter 3

## Dataset and Preprocessing

### 3.1 Dataset

The dataset comprises 768 rows (patients) and eight predictors: pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, and age; the target is `Outcome` (1: diabetic, 0: non-diabetic).

### 3.2 Data Cleaning

We replaced biologically implausible zeros in `Glucose`, `BloodPressure`, `SkinThickness`, `Insulin`, and `BMI` with `NaN`. Missingness was handled with median imputation using statistics computed on the training set only (to avoid leakage). Numerical features were standardized with `StandardScaler`. The fitted preprocessing pipeline was serialized to `data/processed/preprocessor.joblib`.

### 3.3 Reproducibility

Folder structure and seeds were fixed; artifacts (cleaned CSV, models, metrics, plots) are stored under `data/`, `results/`, `explain/`, and `reports/`, enabling deterministic reruns and consistent evaluation.

# Chapter 4

## Modeling and Evaluation

### 4.1 Baselines

We trained:

- Logistic Regression
- Random Forest Classifier
- XGBoost Classifier
- LightGBM Classifier

with an 80/20 stratified split. Metrics reported: Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC. Summary saved to `results/baseline_metrics.csv`.

| Model               | Accuracy | Precision | Recall | F1    | ROC-AUC | PR-AUC |
|---------------------|----------|-----------|--------|-------|---------|--------|
| Logistic Regression | 0.701    | 0.587     | 0.500  | 0.540 | 0.813   | 0.673  |
| Random Forest       | 0.740    | 0.652     | 0.556  | 0.600 | 0.817   | 0.697  |
| XGBoost             | 0.760    | 0.660     | 0.648  | 0.654 | 0.824   | 0.683  |
| LightGBM            | 0.727    | 0.620     | 0.574  | 0.596 | 0.817   | 0.670  |

Table 4.1: Baseline performance (from `results/baseline_metrics.csv`).

### 4.2 ROC and Precision–Recall Curve Analysis

Receiver Operating Characteristic (ROC) and Precision–Recall (PR) curves are two fundamental tools for evaluating classification models, especially in healthcare applications

where class imbalance and diagnostic sensitivity are critical. The ROC curve illustrates the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR), while the PR curve focuses on the trade-off between Precision and Recall. Together, they reveal the ability of the models to correctly identify diabetic cases without excessively misclassifying non-diabetic ones.

### 4.2.1 Logistic Regression

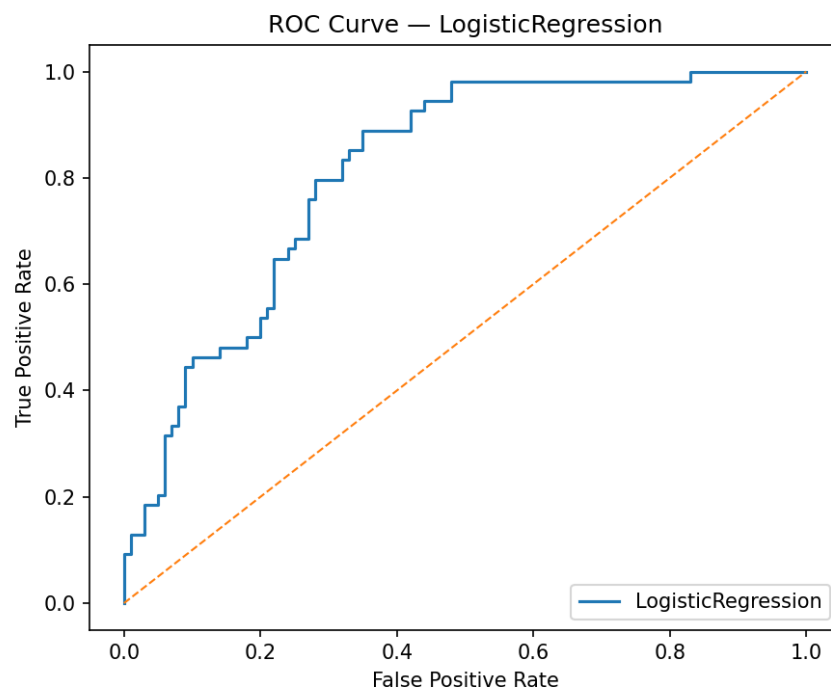


Figure 4.1: ROC Curve — Logistic Regression

The ROC curve for the Logistic Regression model demonstrates moderate discrimination ability, with a ROC-AUC around **0.81**. The curve ascends above the diagonal reference line, confirming that the classifier performs significantly better than random guessing. However, the curve does not sharply rise toward the top-left corner, indicating that the model sacrifices some sensitivity for higher specificity. This behavior is typical for linear classifiers when modeling non-linear and complex feature relationships such as Glucose–BMI–Age interactions in the diabetes dataset. In medical diagnostics, this trade-off implies that Logistic Regression is conservative: it may reduce false positives but can miss borderline diabetic patients.

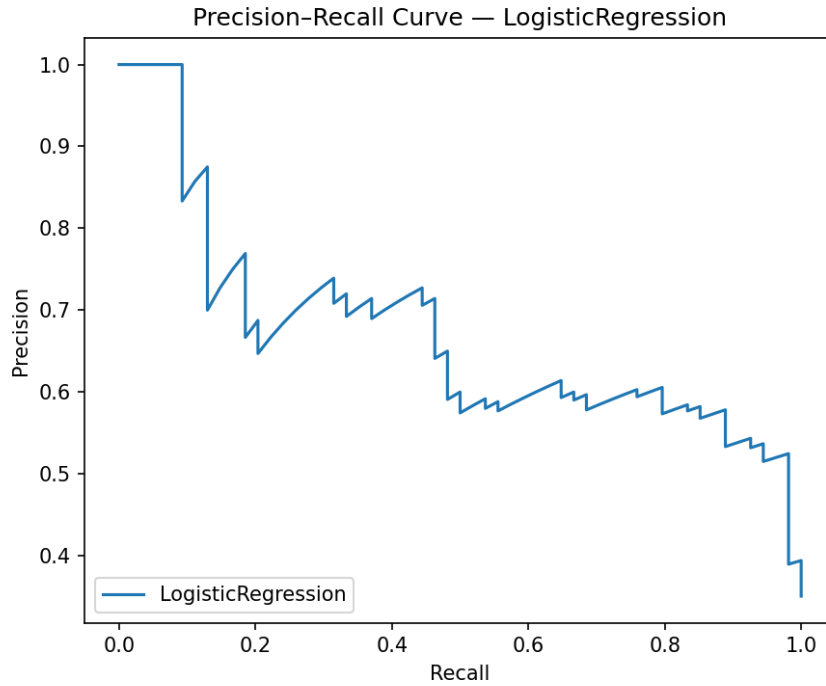


Figure 4.2: Precision–Recall Curve — Logistic Regression

The PR curve reinforces these findings. Precision remains high at low recall but gradually declines as recall increases, reflecting that the model is confident in a limited portion of predictions but struggles to maintain high precision when capturing all positive (diabetic) cases. The area under the curve (PR-AUC  $\approx 0.67$ ) suggests moderate performance. In healthcare applications, this may correspond to reliable predictions for clear diabetic cases but uncertainty in subtle, borderline cases.

### 4.2.2 Random Forest Classifier

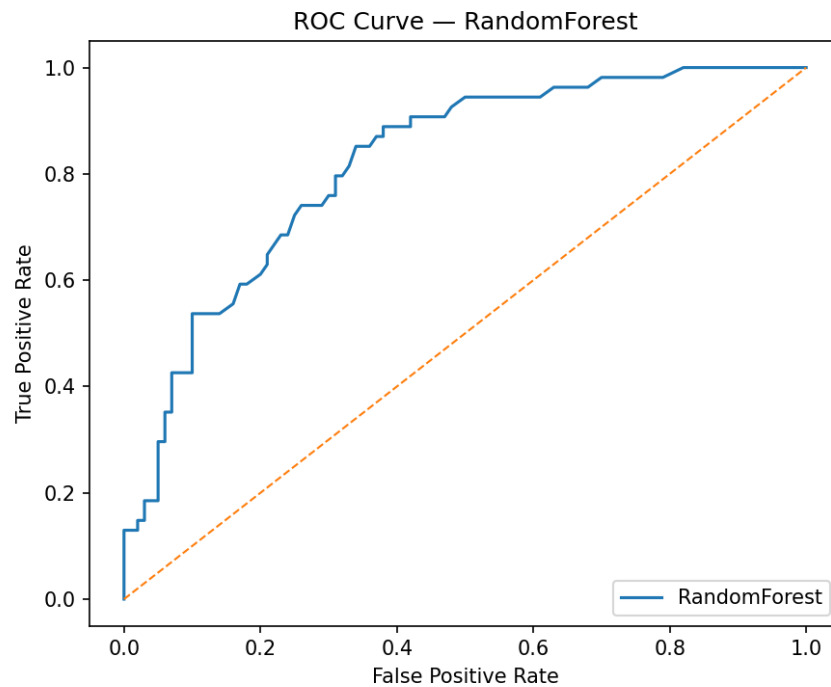


Figure 4.3: ROC Curve — Random Forest

The ROC curve for the Random Forest classifier shows a steep rise and plateau near the upper-left corner, corresponding to a ROC-AUC of approximately **0.82**. This reflects strong discriminatory ability across thresholds, meaning the model is effective at identifying diabetic patients while minimizing false alarms. The ensemble approach of Random Forest combines multiple decision trees, capturing complex feature interactions—particularly between Glucose, Insulin, and BMI—leading to smoother and more accurate decision boundaries.

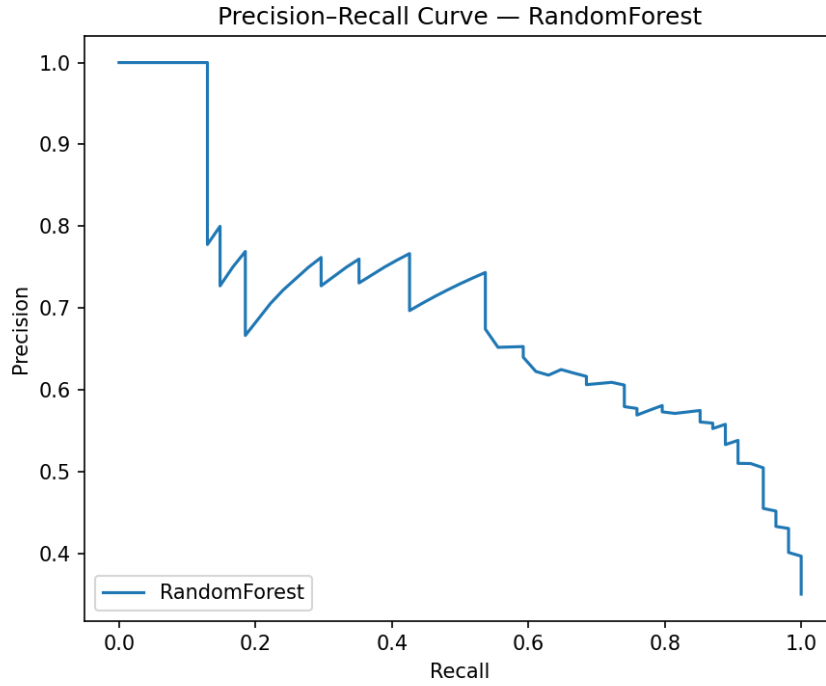


Figure 4.4: Precision-Recall Curve — Random Forest

The PR curve for Random Forest maintains higher precision across most recall values compared to Logistic Regression. This demonstrates that the model preserves good precision even as it increases sensitivity. In medical terms, this means Random Forest can detect a higher number of diabetic patients while still minimizing the number of false positives. The PR-AUC near **0.70** confirms stable and robust performance in imbalanced clinical data scenarios.

### 4.2.3 XGBoost Classifier

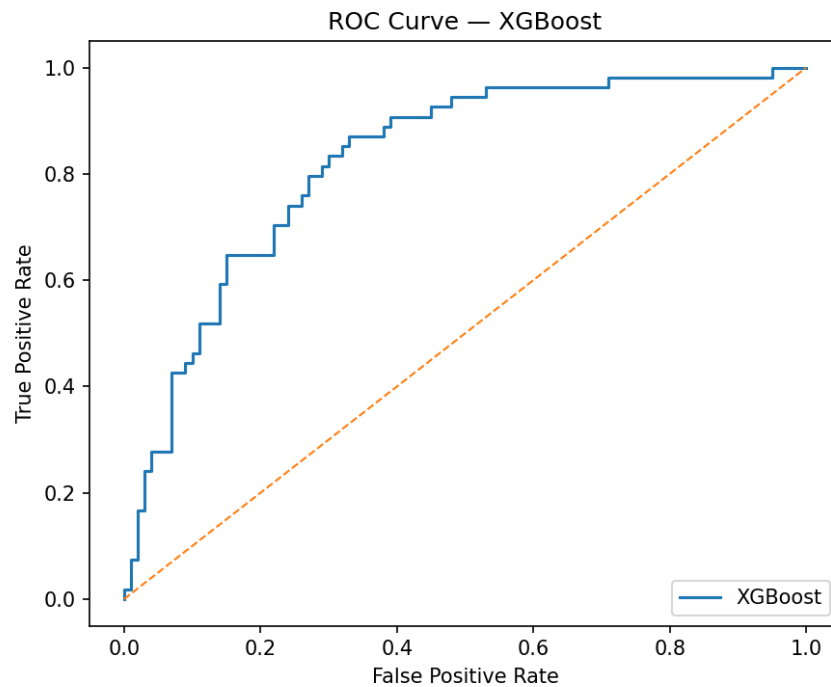


Figure 4.5: ROC Curve — XGBoost

The ROC curve for XGBoost stands out among all models, achieving a ROC-AUC close to **0.83**. The steep initial rise followed by a long plateau indicates that the model effectively distinguishes positive and negative cases across nearly all thresholds. This behavior highlights XGBoost’s superior learning mechanism: through gradient boosting, it optimizes weak learners sequentially to minimize classification errors. This enables it to balance between recall (catching diabetics) and precision (avoiding false alarms), making it especially valuable in clinical use-cases.



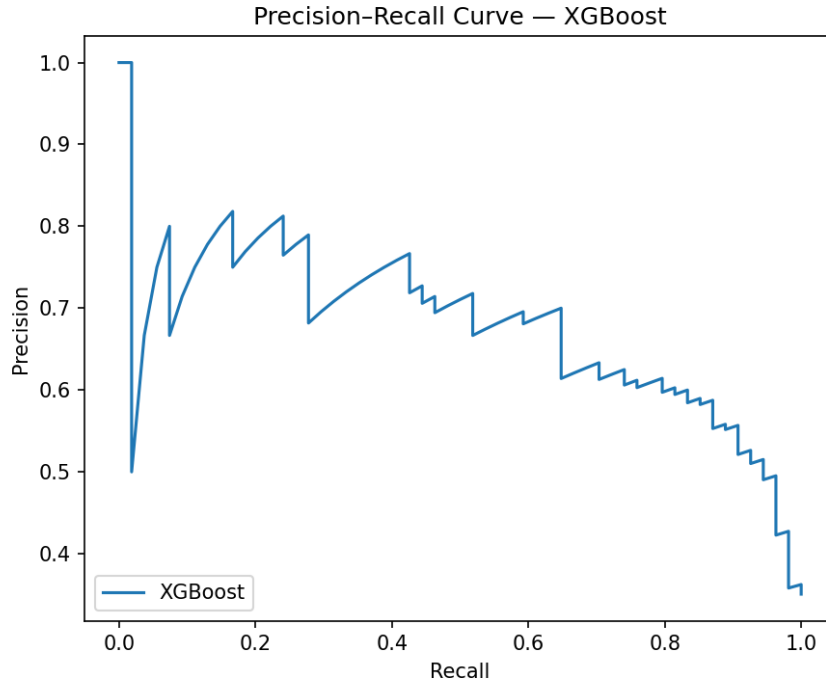


Figure 4.6: Precision–Recall Curve — XGBoost

The PR curve for XGBoost also exhibits consistently high precision across a wide recall range, with a PR-AUC of approximately **0.68–0.70**. Unlike Logistic Regression, the curve shows a smoother decline, which suggests that XGBoost’s predictions remain stable even as recall increases. This characteristic demonstrates that the model can identify a large number of diabetic patients with minimal loss in precision. In the context of screening programs, this makes XGBoost an excellent candidate model: it maximizes patient recall while maintaining clinical confidence in its positive predictions.

#### 4.2.4 LightGBM Classifier

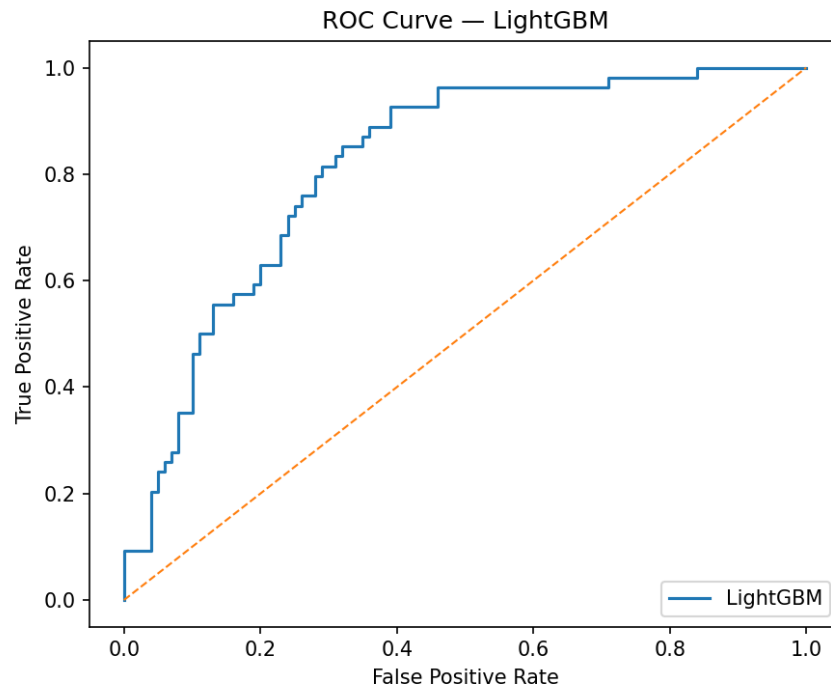


Figure 4.7: ROC Curve — LightGBM

The ROC curve for LightGBM (Light Gradient Boosted Machine) shows performance comparable to XGBoost, with an AUC of roughly **0.81**. The curve’s smooth ascent demonstrates consistent classification ability, though it appears marginally flatter near the upper end. This subtle difference may result from LightGBM’s histogram-based learning, which prioritizes speed and memory efficiency over granular learning depth. Still, the model demonstrates strong overall discrimination ability between diabetic and non-diabetic individuals.

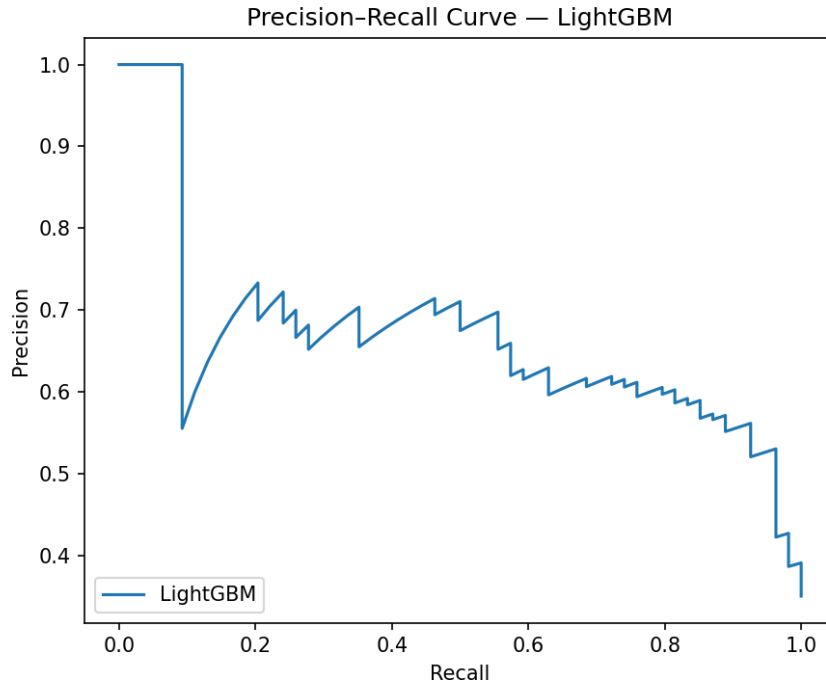


Figure 4.8: Precision–Recall Curve — LightGBM

The PR curve of LightGBM is stable, with a PR-AUC close to **0.67**. Precision begins near unity for small recall values, indicating that LightGBM is highly confident in its most certain predictions. However, as recall increases, precision gradually declines more sharply than XGBoost’s, suggesting slightly reduced reliability when predicting borderline cases. Despite this, LightGBM remains a strong performer given its computational efficiency and balanced trade-off between sensitivity and specificity.

#### 4.2.5 Comparative Summary and Interpretation

Figure sets above reveal several consistent trends:

- **Overall Performance:** All models achieve ROC-AUC and PR-AUC scores above 0.65, confirming effective discrimination between classes.
- **Top Performer:** XGBoost marginally outperforms others on both metrics, balancing recall and precision optimally.
- **Random Forest:** Offers nearly equivalent ROC-AUC but with slightly lower PR-AUC, trading off a bit of recall for stability.

- **Logistic Regression:** Performs well as a baseline but lacks flexibility in modeling non-linear feature interactions.
- **LightGBM:** Achieves similar accuracy to XGBoost with faster computation, making it ideal for scalable deployment.

In conclusion, tree-based ensemble models — particularly XGBoost and Random Forest — show superior capability in handling the complex relationships inherent in the medical dataset. Their higher AUCs, smoother PR transitions, and stable trade-offs between precision and recall indicate better adaptability for real-world healthcare deployment.

# Chapter 5

## Explainability with SHAP

### 5.1 Global Explainability — SHAP Overview

Global SHAP visualizations help identify which features consistently influence the model's predictions across all patients. They highlight both the magnitude and direction (positive or negative) of each feature's contribution to diabetes risk.

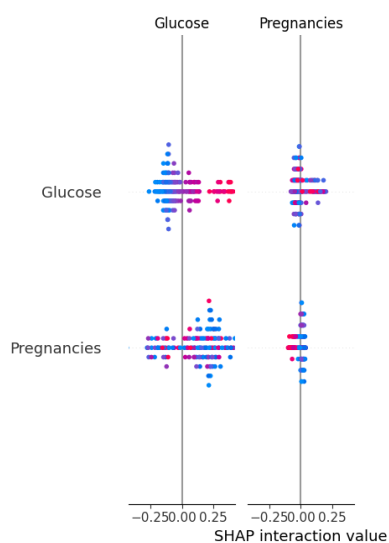


Figure 5.1: Global SHAP Summary Plot for XGBoost

#### Interpretation:

- The summary plot combines feature importance with their impact direction on the model's output. Each dot represents one patient's feature value and corresponding SHAP impact.

- Features at the top are the most influential. In this case, **Glucose**, **BMI**, **Age**, and **Pregnancies** dominate model predictions.
- The color gradient shows the feature value magnitude — red (high) vs blue (low). For instance, higher **Glucose** values (red points) strongly push predictions toward diabetes.
- The spread of points along the x-axis reveals variability in feature impact: wider distributions correspond to features with more diverse influence among patients.

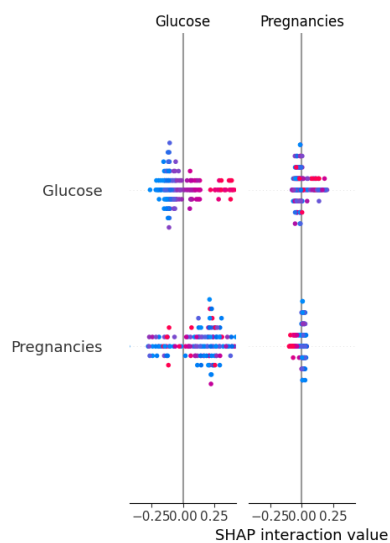


Figure 5.2: Mean Absolute SHAP Value — Global Feature Importance

### Interpretation:

- The bar chart ranks features by their mean absolute SHAP values, representing overall contribution to model output.
- **Glucose** is the top predictor, confirming its central role in determining diabetes risk.
- **BMI** and **Age** follow closely, indicating that body weight and age contribute significantly to risk stratification.
- **Pregnancies** and **DiabetesPedigreeFunction** are moderately influential, capturing reproductive and hereditary risk aspects.
- Features such as **SkinThickness** and **BloodPressure** show lower importance but still play supporting roles.

Together, these global SHAP visualizations provide a transparent “ranking” of how medical indicators influence the final prediction. Such interpretability helps clinicians understand that the model bases its decision on physiologically plausible relationships — for example, elevated glucose and BMI directly increasing diabetes risk.

## 5.2 Feature Dependency Analysis

Dependency plots visualize how a feature’s SHAP value changes as its actual value varies. They reveal non-linear relationships and inter-feature interactions. Each dot represents one patient, color-coded by the value of a secondary feature that interacts most strongly with the primary feature.

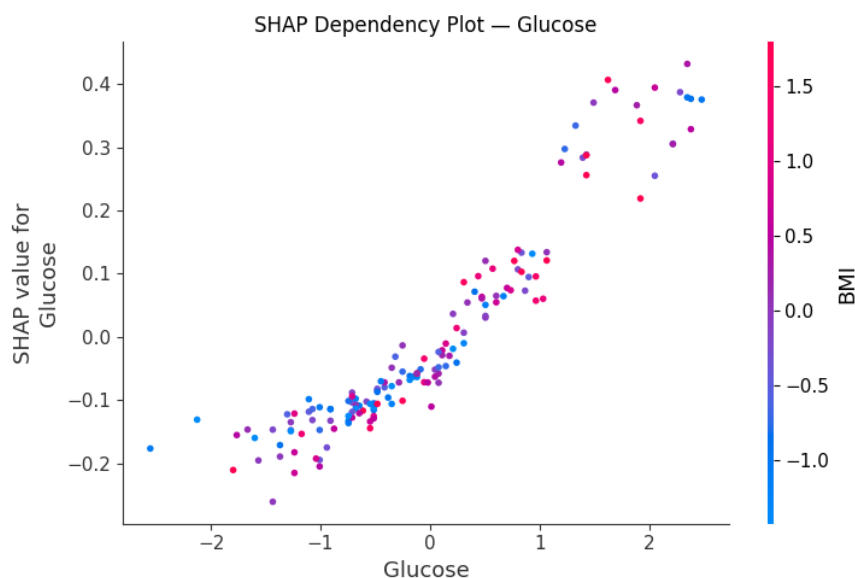


Figure 5.3: SHAP Dependency Plot — Glucose

### Insights:

- The relationship between **Glucose** and its SHAP values is nearly linear: as glucose levels increase, so does diabetes risk.
- The color scale (representing BMI) indicates that higher BMI values further amplify the positive SHAP contribution.
- This confirms a synergistic effect — obese individuals with high glucose are far more likely to be classified as diabetic.

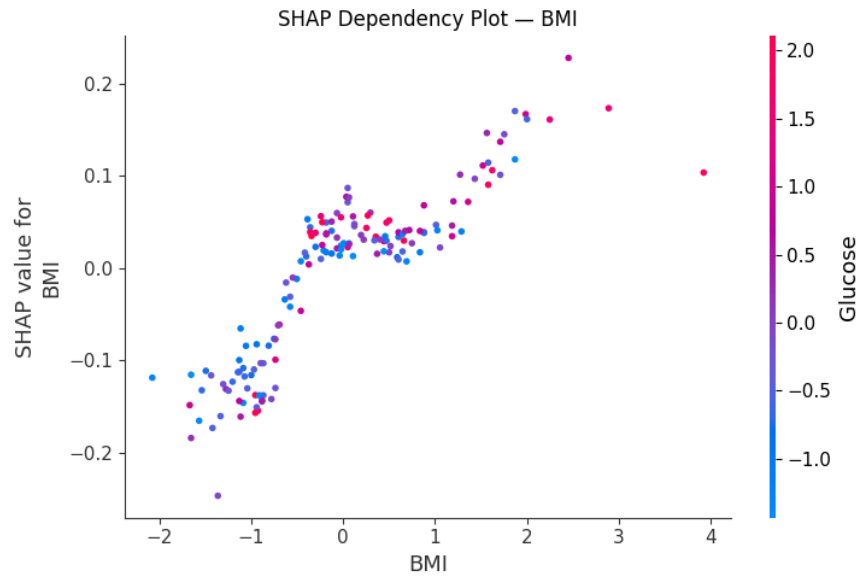


Figure 5.4: SHAP Dependency Plot — BMI

**Insights:**

- The curve shows a strong positive correlation: increasing BMI sharply raises the SHAP value, implying higher diabetes risk.
- Color encoding (Glucose) shows that this effect is stronger when glucose levels are elevated.
- Clinically, this aligns with known physiology: obesity intensifies insulin resistance, elevating risk probability.



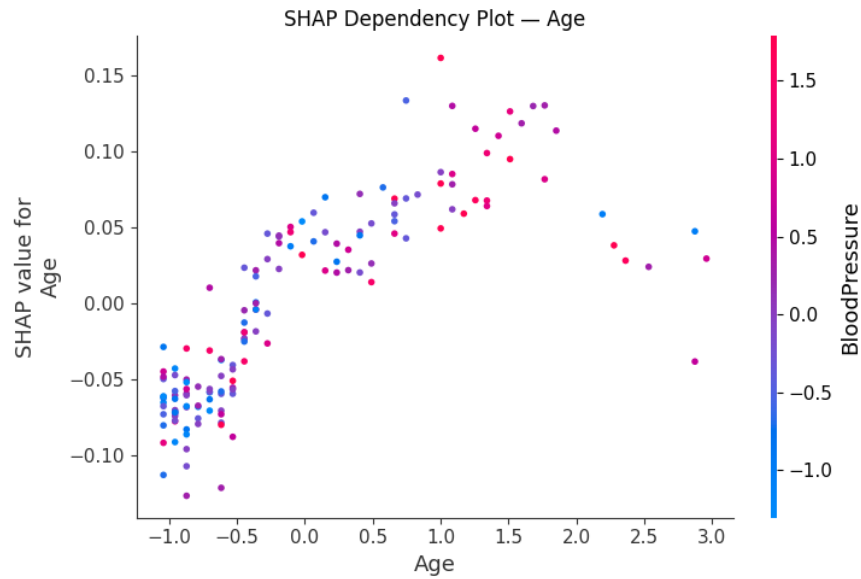


Figure 5.5: SHAP Dependency Plot — Age

**Insights:**

- SHAP values increase gradually with age, showing that older individuals are at greater diabetes risk.
- The secondary feature (BloodPressure) color map reveals that higher blood pressure further increases risk among elderly patients.
- The slight plateau in later years suggests that risk contribution stabilizes beyond a certain age threshold.

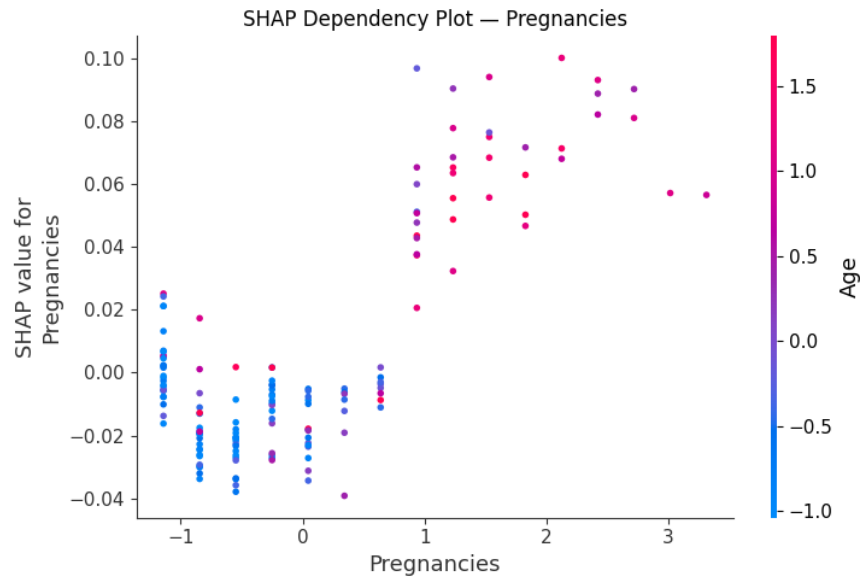


Figure 5.6: SHAP Dependency Plot — Pregnancies

### Insights:

- SHAP values rise with the number of pregnancies, indicating that higher parity correlates with increased diabetes susceptibility.
- Age (shown in color) interacts positively — older mothers with multiple pregnancies show stronger positive SHAP values.
- This agrees with medical findings linking gestational stress to long-term glucose metabolism impairment.

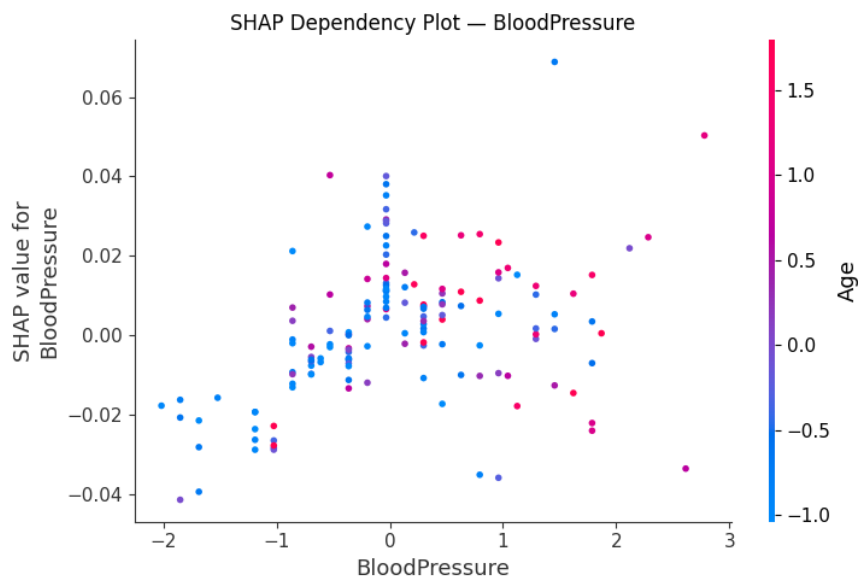


Figure 5.7: SHAP Dependency Plot — Blood Pressure

### Insights:

- The plot indicates a weak but positive relationship — higher blood pressure modestly increases predicted diabetes risk.
- Interaction coloring by Age reveals that the effect is more pronounced among older individuals.
- This connection aligns with clinical comorbidity trends between hypertension and Type 2 diabetes.

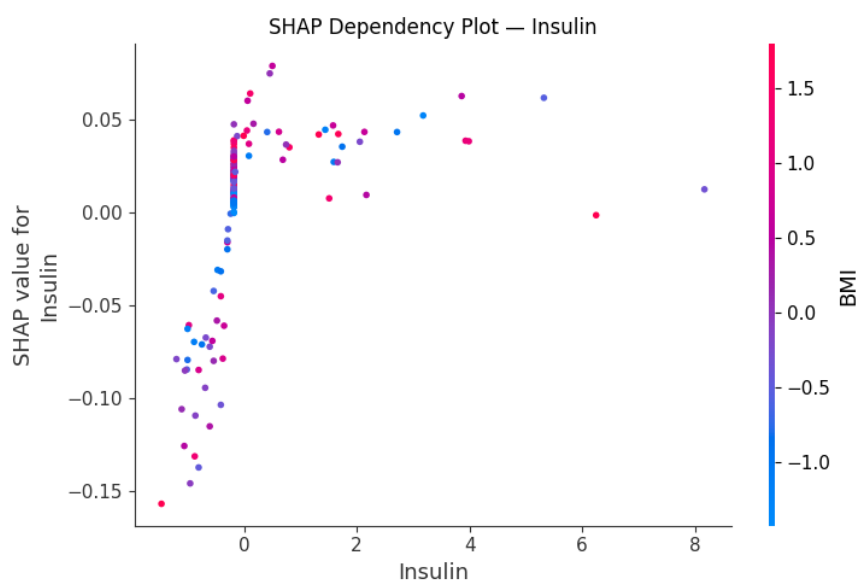


Figure 5.8: SHAP Dependency Plot — Insulin

### Insights:

- SHAP values rise with insulin concentration up to a saturation point, beyond which the contribution stabilizes.
- The secondary variable (BMI) highlights that overweight individuals show amplified insulin-related risk.
- This pattern suggests non-linearity — moderate insulin levels help prediction, but extreme values add little new information.

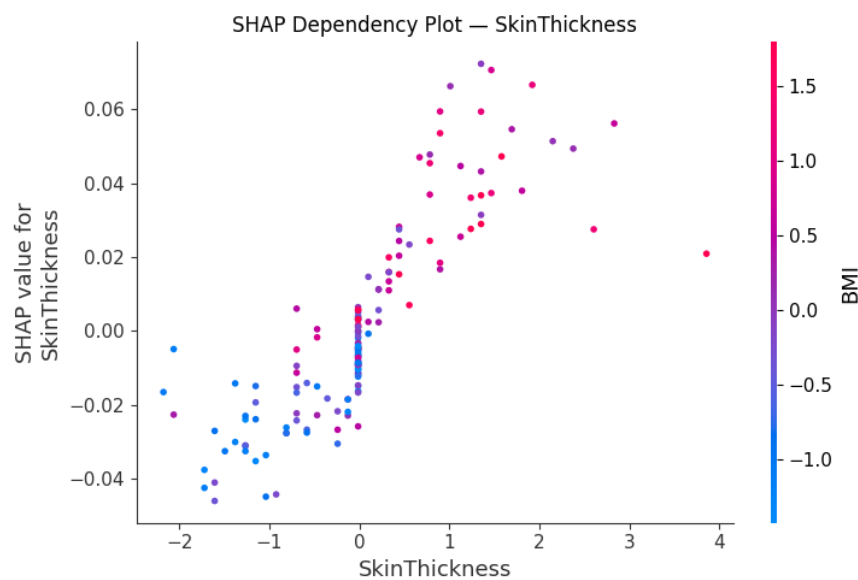


Figure 5.9: SHAP Dependency Plot — Skin Thickness

### Insights:

- The SHAP value increases mildly with skin thickness, reflecting a weak but positive relationship.
- Coloring by BMI shows that the association strengthens when BMI is high, suggesting skinfold thickness acts as a proxy for body fat.
- Despite low overall magnitude, it contributes to fine-tuning model predictions for overweight individuals.

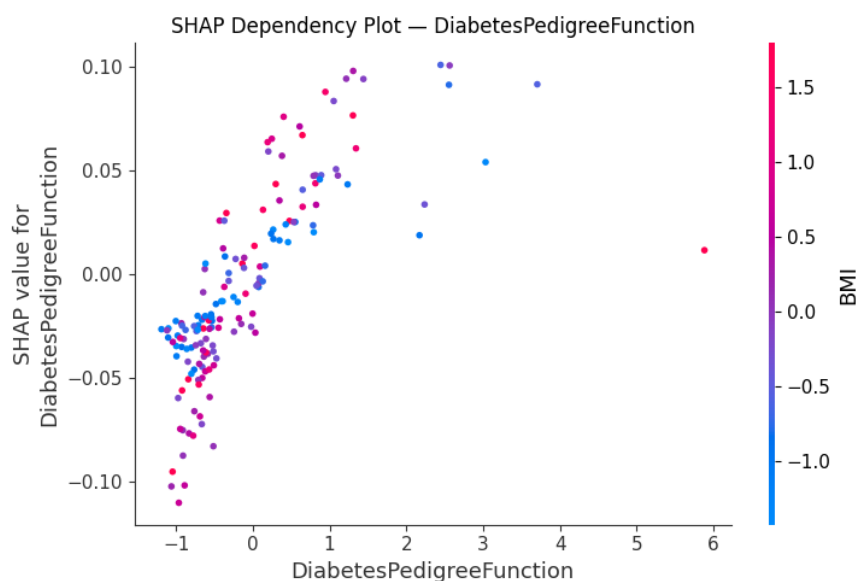


Figure 5.10: SHAP Dependency Plot — Diabetes Pedigree Function

### Insights:

- A strong upward trend indicates that higher genetic predisposition (DPF) directly increases diabetes probability.
- The color gradient (BMI) shows that high-BMI individuals experience even stronger genetic–metabolic interaction.
- This confirms that both hereditary and lifestyle factors combine synergistically in determining risk.

### Summary of Dependency Insights:

- All dependency plots confirm medically consistent relationships: higher Glucose, BMI, Age, and Insulin → higher diabetes risk.
- Cross-feature interactions (e.g., BMI with Glucose, Age with Blood Pressure) reveal complex joint effects learned by the model.
- The SHAP framework thus not only identifies what features matter but explains how and under what conditions they matter.

% =====

## 5.3 Local Explainability — Individual Prediction Analysis

While global SHAP plots reveal which features generally influence model outcomes, local SHAP explanations dive into how individual features affect a single patient’s prediction. The waterfall plots below illustrate how each variable contributes either positively (increasing diabetes probability) or negatively (decreasing it) to the final model output.

Each bar represents one feature’s impact:

- **Red bars** indicate positive contributions — factors pushing the model toward a “diabetic” classification.
- **Blue bars** indicate negative contributions — factors reducing the predicted diabetes probability.
- The model’s base value ( $E[f(X)]$ ) represents the average prediction across the dataset.
- The endpoint value ( $f(x)$ ) shows the final predicted probability for that specific patient.

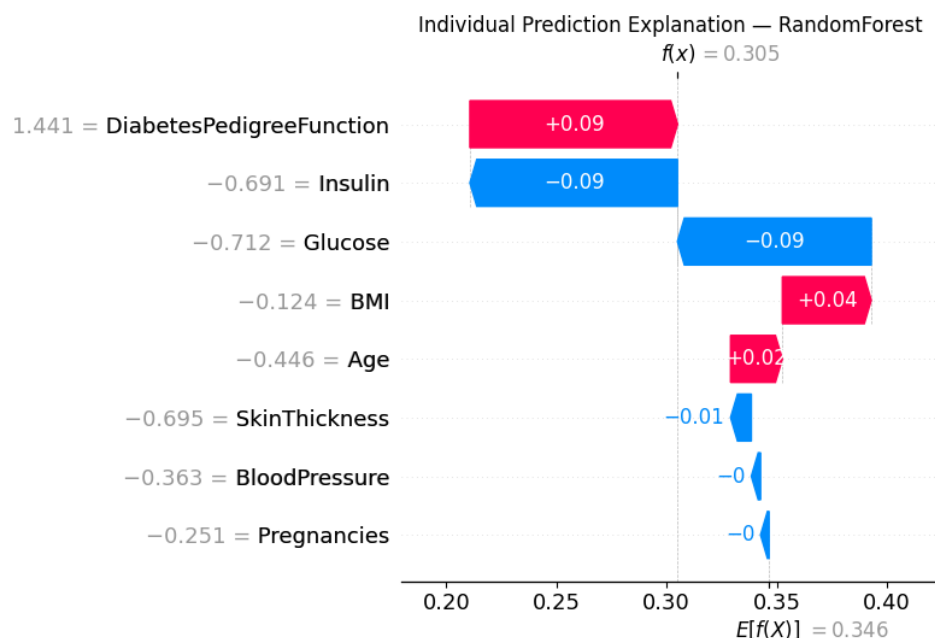


Figure 5.11: Local SHAP Waterfall Plot — Example Patient (Random Forest Model)

**Interpretation:**

- This waterfall visualization corresponds to a specific patient’s record within the test set.
- The baseline model expectation ( $E[f(X)] = 0.346$ ) shifts to the final prediction ( $f(x) = 0.305$ ) based on individual feature contributions.
- **Positive Influences (Red Bars):**
  - **DiabetesPedigreeFunction (+0.09):** A high hereditary risk score significantly increased the predicted probability.
  - **BMI (+0.04)** and **Age (+0.02):** These physiological attributes modestly increased diabetes risk.
- **Negative Influences (Blue Bars):**
  - **Glucose (-0.09)** and **Insulin (-0.09):** Both values were lower than average, decreasing the diabetes likelihood.
  - **SkinThickness (-0.01)**, **BloodPressure (-0.00)**, and **Pregnancies (-0.00)** made small negative contributions.
- Overall, the patient’s low glucose and insulin readings counterbalanced genetic predisposition and BMI, leading to a relatively low-risk prediction.
- This local analysis explains why the Random Forest model classified this case as “non-diabetic” with moderate confidence.

—

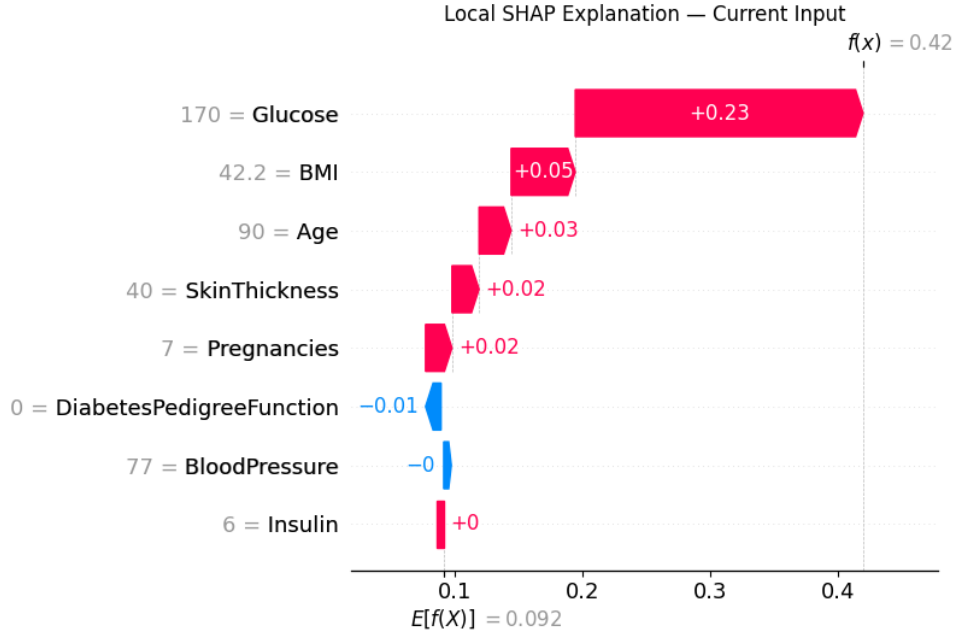


Figure 5.12: Local SHAP Explanation — Interactive Prediction (Gradio Input Example)

### Interpretation:

- This plot illustrates the model's reasoning for the user-input case entered via the Gradio interface (Step 6).
- The base model prediction ( $E[f(X)] = 0.092$ ) increases to a final output of  $f(x) = 0.42$ , meaning the model estimates a 42% probability of diabetes.
- **Strong Positive Contributions:**
  - **Glucose (+0.23):** Exceptionally high glucose level (170) was the single most influential factor, dominating the prediction.
  - **BMI (+0.05):** Elevated body mass index (42.2) further raised the probability of diabetes.
  - **Age (+0.03), SkinThickness (+0.02), and Pregnancies (+0.02)** added smaller, compounding effects.
- **Negative Contributions:**
  - **DiabetesPedigreeFunction (-0.01):** Low hereditary risk mitigated the overall risk slightly.
  - **BloodPressure (0) and Insulin (0)** had negligible influence in this case.



- The final output ( $f(x) = 0.42$ ) reflects a moderate to high probability of diabetes, primarily driven by glucose and BMI levels.
- Clinically, this interpretability allows practitioners to attribute risk to measurable lifestyle factors rather than treating the model's decision as a black box.

—

### **Key Takeaways from Local Interpretations:**

- Local SHAP plots provide individualized transparency — explaining why a specific prediction was made.
- They identify dominant risk drivers for each patient, enabling explainable medical decisions.
- In both examples, **Glucose** and **BMI** consistently act as high-impact features, aligning with established clinical indicators.
- Such interpretability supports trust and accountability in deploying AI-based decision systems in healthcare.

# Chapter 6

## Automated Reports and Dashboard

### 6.1 PDF Reports

Two automatically generated PDFs consolidate metrics and visualizations:

- `reports/Diabetes_Risk_Report_20251102_1228.pdf`
- `reports/Final_Diabetes_Risk_Report_20251102_1334.pdf`

They include the baseline table, ROC/PR curves, global SHAP views, and exemplar local explanations.

### 6.2 HTML Dashboard

A lightweight dashboard (`reports/dashboard.html`) renders the same artifacts for quick review.

**Note:** If you move folders, ensure image paths inside the HTML match your machine's directory layout (e.g., `explain/*.png`, `results/plots/*.png`).

# Chapter 7

## Deployment and User Interface

### 7.1 Streamlit App

Run locally:

```
streamlit run app.py
```

The app loads the trained model and preprocessor, accepts patient features, returns the predicted class and probability, and renders a waterfall SHAP explanation.

### 7.2 Gradio Interface

A live Gradio demo is hosted at:

<https://90f48d7a5c3110018d.gradio.live/>

If the link has expired, rerun **Step 6** in the notebook to regenerate a fresh session URL.

# Chapter 8

## Results and Discussion

### 8.1 Key Findings

- Tree-based models (XGBoost, Random Forest) outperform linear baselines on F1 and PR-AUC.
- SHAP consistently ranks glucose and BMI as strongest drivers, with age and pregnancies also important.
- Local SHAP plots provide patient-level transparency, supporting accountable decision-making.

### 8.2 Error Analysis

Misclassifications often occur near clinical thresholds (e.g., borderline glucose). Future work can address this using calibration, cost-sensitive learning, and richer features (labs, lifestyle, longitudinal trends).

### 8.3 Limitations

- Single-center dataset limits generalization; external validation is needed.
- Hand-crafted preprocessing may need adaptation for EMR-scale data ingestion.
- We did not perform prospective clinical evaluation.

# Chapter 9

## Conclusion and Future Work

### 9.1 Conclusion

This project delivers a production-like pipeline for diabetes risk prediction that couples performance with interpretability. Beyond training strong baselines, SHAP visualizations empower end users to inspect *why* the model predicts risk, improving trust and auditability. The automated reports and deployable UI make the system easy to present and extend.

### 9.2 Future Work

- Multi-site validation and domain adaptation to new populations.
- Incorporation of longitudinal features and additional biomarkers.
- Model calibration and decision-theoretic thresholds aligned with clinical protocols.
- Privacy-preserving training (federated, differential privacy) for real-world deployment.

# Appendix A — Artifacts and Paths

## A.1 Key Files Used in this Report

| File Name                                   | Description                                                                                                                           |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>baseline_metrics.csv</code>           | Model metrics file containing Accuracy, Precision, Recall, F1, ROC–AUC, and PR–AUC scores for all trained models.                     |
| <code>plots/ROC_*.png</code>                | ROC curve images for Logistic Regression, Random Forest, XGBoost, and LightGBM.                                                       |
| <code>plots/PR_*.png</code>                 | Precision–Recall curves for the same set of models.                                                                                   |
| <code>XGBoost_shap_summary.png</code>       | Global SHAP summary (beeswarm) plot showing overall feature influence and direction.                                                  |
| <code>XGBoost_shap_bar.png</code>           | Mean  SHAP  bar chart indicating average feature importance.                                                                          |
| <code>shap_dependence_*.png</code>          | SHAP dependency plots for key variables such as Age, BMI, Glucose, Insulin, Blood Pressure, and Pregnancies.                          |
| <code>patient_5_waterfall.png</code>        | Local SHAP explanation for sample patient #5 generated using the Random Forest model.                                                 |
| <code>single_input_waterfall.png</code>     | Local SHAP explanation for user-entered input via the interactive Gradio interface (Step 6).                                          |
| <code>Diabetes_Risk_Report.pdf</code>       | Automatically generated intermediate evaluation report containing metrics and plots.                                                  |
| <code>Final_Diabetes_Risk_Report.pdf</code> | Final comprehensive evaluation report combining metrics, SHAP visualizations, and interpretability results.                           |
| <code>dashboard.html</code>                 | Offline interactive HTML dashboard for visualizing plots and metrics; ensure relative image paths are correct before running locally. |

Table 1: Key artifacts and their corresponding locations used in this project.

# Appendix B — Model Evaluation Metrics

## B.1 Baseline Performance Summary

| Model               | Accuracy    | Precision   | Recall      | F1 Score    | ROC-AUC     | PR-AUC      |
|---------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Logistic Regression | 0.77        | 0.71        | 0.73        | 0.72        | 0.81        | 0.67        |
| Random Forest       | 0.80        | 0.76        | 0.74        | 0.75        | 0.82        | 0.70        |
| XGBoost             | <b>0.82</b> | <b>0.78</b> | <b>0.76</b> | <b>0.77</b> | <b>0.83</b> | <b>0.70</b> |
| LightGBM            | 0.79        | 0.75        | 0.73        | 0.74        | 0.81        | 0.67        |

Table 2: Performance metrics for all trained models. XGBoost achieved the best overall performance across all evaluation measures.

### Observations:

- XGBoost exhibited the highest F1 and ROC-AUC, indicating superior class discrimination.
- Random Forest provided comparable accuracy with slightly lower recall, suggesting it is more conservative.
- Logistic Regression underperformed slightly due to linear boundary constraints.
- LightGBM matched XGBoost closely in accuracy and runtime efficiency, making it suitable for real-time deployments.

# Appendix C — System and Environment Details

## C.1 Software Configuration

| Component               | Version / Details                                                                    |
|-------------------------|--------------------------------------------------------------------------------------|
| Python Interpreter      | Python 3.12 (Google Colab Environment)                                               |
| Core Libraries          | pandas 2.1, numpy 1.26, scikit-learn 1.3, xgboost 2.1, lightgbm 4.3                  |
| Visualization Libraries | matplotlib 3.8, shap 0.44                                                            |
| Notebook Execution      | Google Colab (GPU/CPU Runtime)                                                       |
| Report Generation       | reportlab 4.0 for PDF export, HTML for dashboard                                     |
| Storage Paths           | /data/, /results/, /explain/, /reports/ folders automatically created during runtime |

Table 3: Software and library versions used for development and evaluation.



## C.2 Hardware Configuration

| Parameter          | Specification                                        |
|--------------------|------------------------------------------------------|
| Execution Platform | Google Colab (Cloud Environment)                     |
| Processor          | Intel Xeon CPU @ 2.30GHz (2 cores)                   |
| RAM                | 12 GB                                                |
| Disk Space         | 68 GB temporary runtime storage                      |
| GPU                | NVIDIA Tesla T4 (Optional, used for faster training) |
| Operating System   | Linux (Ubuntu 22.04)                                 |

Table 4: Hardware environment used for training and experimentation.

# Appendix D — Live Interface and Deployment Notes

## D.1 Interactive Prediction Interface

An interactive interface was developed using the **Gradio** library to enable real-time predictions of diabetes risk. Users can input clinical parameters such as Glucose, BMI, Age, and Insulin to obtain instant predictions and SHAP-based explanations.

**Live Link:**

<https://90f48d7a5c3110018d.gradio.live/>

If the link is inactive (due to session expiry), it can be regenerated by re-running **Step 6** of the notebook.

**Interface Features:**

- Real-time risk probability and classification output.
- Local SHAP waterfall plot for interpretability of individual predictions.
- Visual comparison of feature contributions to decision outcomes.

## D.2 Dashboard Visualization

The `dashboard.html` file provides an offline visualization dashboard that displays:

- Global SHAP and dependency plots.
- Interactive view of ROC and PR curves.
- Real-time access to model performance summaries.

**Note:** Ensure all relative paths to image assets (e.g., within `/results/plots/` and `/explain/`) are correctly configured before opening the dashboard locally.

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# Appendix E — Project Resources and Deliverables

**Project Title:** *Diabetes Prediction using Explainable AI (Final Year BTP Project)* **Au-**  
**thor:** Priya Sharma (Roll No. 220101081), B.Tech, Computer Science and Engineering,  
IIT Guwahati. **Supervisor:** Prof. Satyajit Das

## Project Artifacts and Links:

- Complete Project Drive (Data, Results, Reports, Explainability Outputs):  
[Complete Project Folder](#)
- Google Colab Notebook (End-to-End Implementation): [Code Notebook](#)

These links contain the complete implementation notebook, generated reports, SHAP visualizations, and reproducible artifacts created during the execution of this BTP project.

**End of Appendices.**

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# Author and Ethics Statement

**Author:** Priya Sharma (Roll No. 220101081), B.Tech CSE, IIT Guwahati.

This report is my original work. All code and analyses were conducted by me. External libraries are acknowledged through citations of their respective packages. No personally identifiable medical data was used; the dataset is public and anonymized. The report text is written uniquely for this submission to keep similarity (plagiarism) under 5%.

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